GHAZI UNIVERSITY
DERA GHAZI KHAN

PF LAB

Name	:	M.Usman
Program	:	BS CS
Section	:	C
Roll Number	:	07



CamScanner

Date: 09-10-2026

★★★

Date:_____

★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★★

Task # 02

```
# include <iostream>

int main ( )
{
    int num1, num2 ;

cout << "Enter first integer" : " ;
    cin >> num ;
cout << "Enter second integer: " ;
    cin >> num2;
 cout << "Addition = " << num1 + num2 << endl;
cout << "subtraction =" << num1 - num2 << endl;
cout << " Multiplication = " << num1 * num2 << endl;
cout << " Division = " << num1 / num2 << endl;

    return 0;

}
```

output

↳ Enter first integer.

next

Task # 03

```
#include <iostream>

int main ( )
{
    int val1;
    int val2;
    int val3;

cout << "please enter your 3 numbers:";
 cin >> val1 >> val2 >> val3;

cout << val1 << "\n" << val2 << "\n"
   << val3 << "\n";

cout << val3 << "\n" << val2 << "\n"
   << val1 << "\n";
return 0;
}
```

output →

please enter your 3 Numbers

like

1
2
3
3
2
;

next

Date: _____

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Task # 04

```
# include <iostream>
# include <cmath>
    int main ( )
    {
      double x1 , y1 , x2 , y2 , distance;
cout << "Enter x1: ";
  cin >> x1;
cout << "Enter y1: ";
  cin >> y1;
cout << "Enter x2: ";
  cin >> x2;
cout << "Enter y2: ";
  cin >> y2;
distance = sqrt((x2-x1) * (x2-x1)+(y2-y1)
  * (y2-y1));
cout << "Distance between the two
  points = " << distance << endl;
  return 0;
  }
```

output

[Enter x1]

next

Date:_____

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Task # 05

```
#include <iostream>
int main()
{
double radius, length, width, base
height;
double circleArea, rectangularArea, tri-Area;
const double PI = 3.14159;
//circle
cout << "Enter radius of circle: ";
cin >> radius
circleArea = PI * radius * radius;
// Rectangular
cout << "Enter length of rectangle; ";
cin >> length;
cout << "Enter width of rectangle: ";
cin >> width;
cout Rectangular Area = length * width;
// Triangle
cout << Enter base of triangle: ";
cin >> base;
```

next

Date:_____

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Task # 06

```
# include <iostream>

int main ( )
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
  double subtotal, salestax, total;
 subtotal = item1 + item2 + item3 + item4 + item5;
   sales tax = subtotal * 0.06;
    total = subtotal + saletax;
   cout << "price of item 1 = $" << item1 << endl;
cout << "price of item2 = $" << item2 << endl;
cout << "price of item3 = $" << item3 << endl;
cout << "Price of item4 = $" << item4 << endl;
cout << "Price of item5 = $ << item5 << endl;

cout << "subtotal = $" << subtotal << endl;
cout << "sales tax (6%) = $" << salestax << endl;
 cout << "total = $" << total << endl;
    return 0;
}
```

next

Date:_____

#06

output

price of item 1 = $ 12.95

price of item2 = $ 24.95

price of item3 = $ 6.95

price of item4 = $ 14.95

price of item 5 = $ 3.95

subtotal = $ 63.75

Sales Tax (6%) = $ 3.825

total = $ 67.575

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#06

output

price of item 1 = $ 12.95

price of item2 = $ 24.95

price of item3 = $ 6.95

price of item 4 = $ 14.95

price of item 5 = $ 3.95

subtotal = $ 63.75

Sales Tax (6%) = $ 3.825

total = $ 67.575



Date:_____

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Bonus Task

```
# include <iostream>
    int main ( )
{
    int num;

cout << "Enter the five digit number:";
    cin >> num;
int digit 1 = num / 10000;
int digit 2 = (num / 1000) % 10;
int digit 3 = (num / 100) % 10;
int digit 4 = (num / 10) % 10;
int digit 5 = num % 10;

cout << digit 1 << "  "
   << digit 2 << "  "
   << digit 3 << "  "
   << digit 4 << "  "
   << digit 5 << "  "
    return 0;
}
```

output

[Enter the five digit number]

next

Date: _____

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Home Task

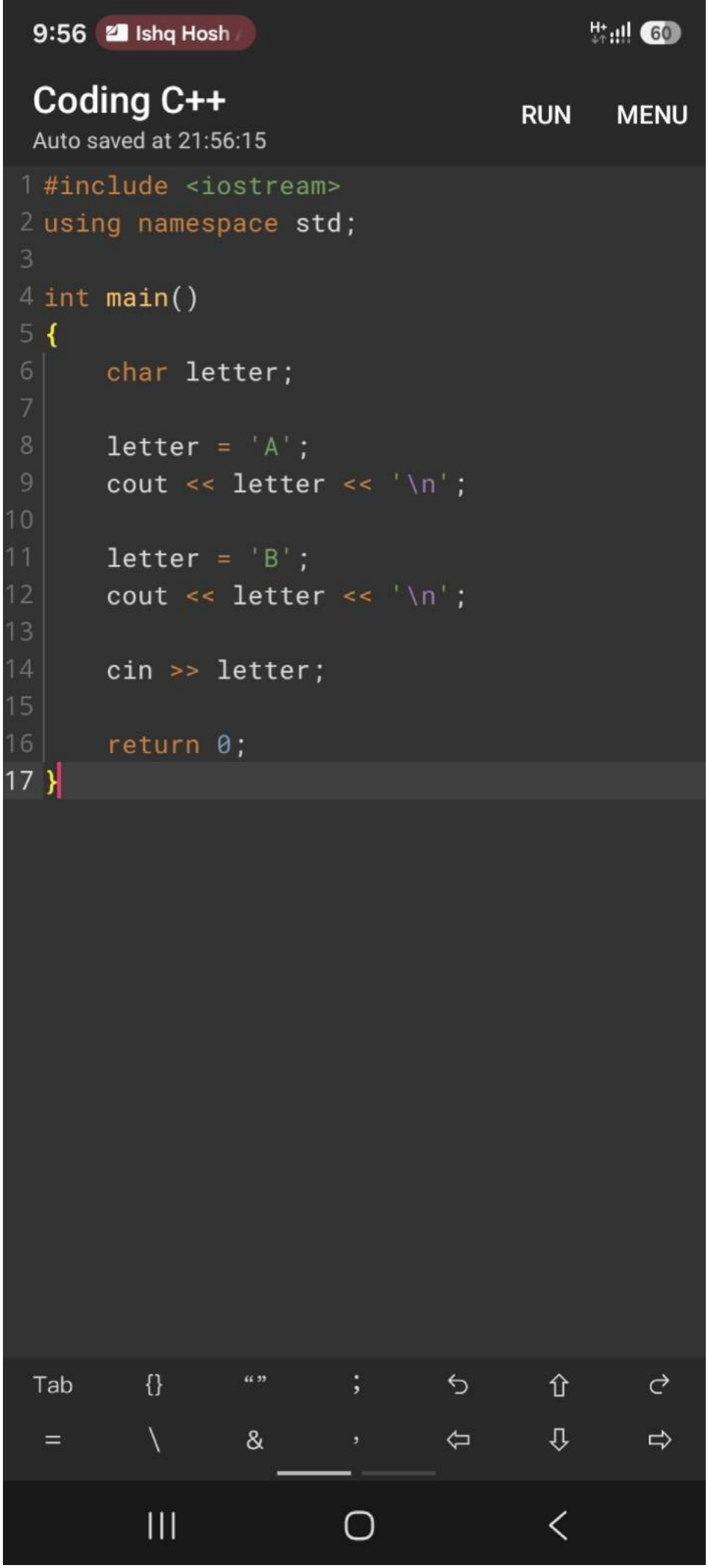
```
#include <iostream>
    int main ( )
{
{
        int var1, var2, var3, temp;
        cin >> var1 ;
cout << "Enter value of var1 : ";
    cin >>  var1 ;
cout << "Enter value of var2: " ;
    cin >>  var2 ;
    temp =  var1 ;
    var1 =  var2;
    var2 =  Temp ;
cout << "After swapping:" << endl;
Cout << " var1 = " << var1 << endl;
Cout << " var2 = " << var2 << endl;

    return 0;
}
}
```

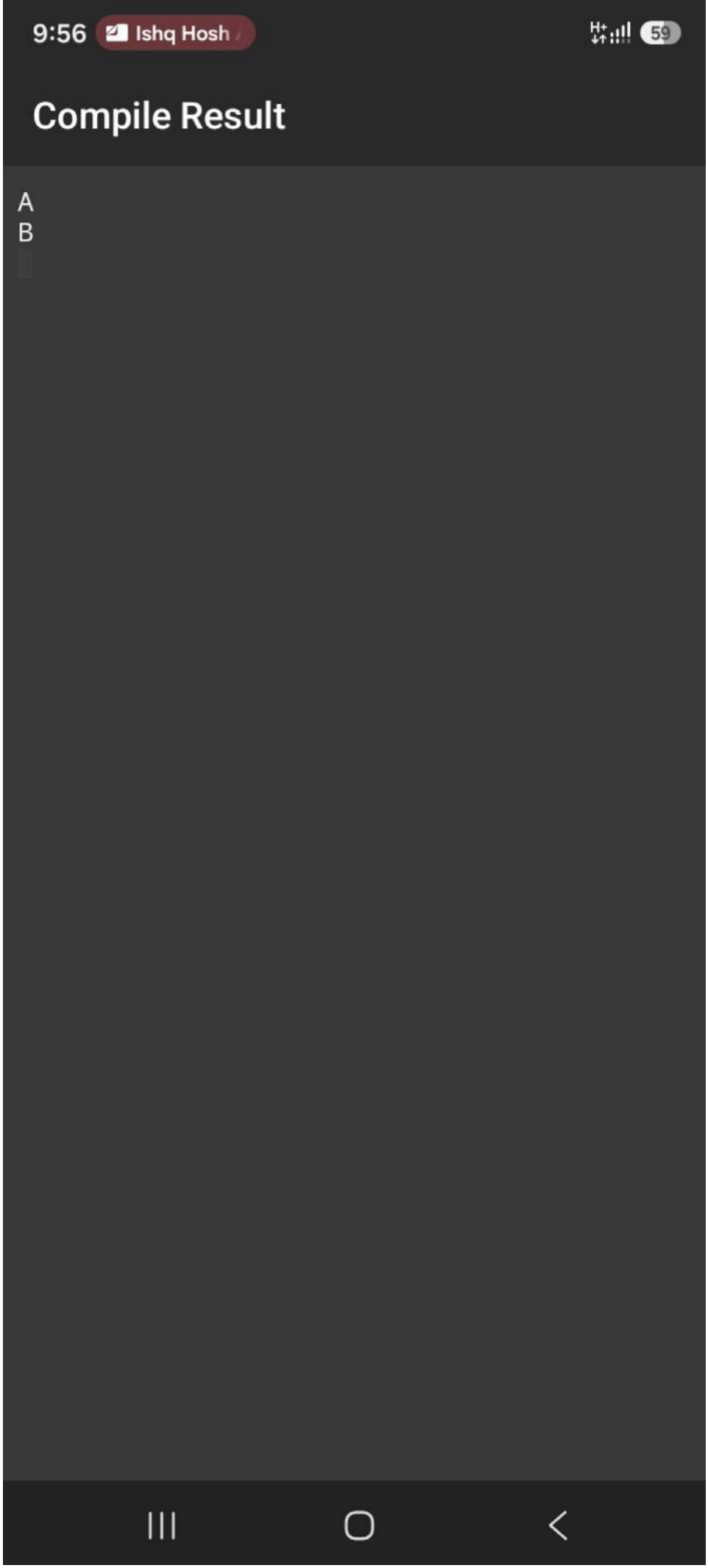
output

↳ Enter the value of var1

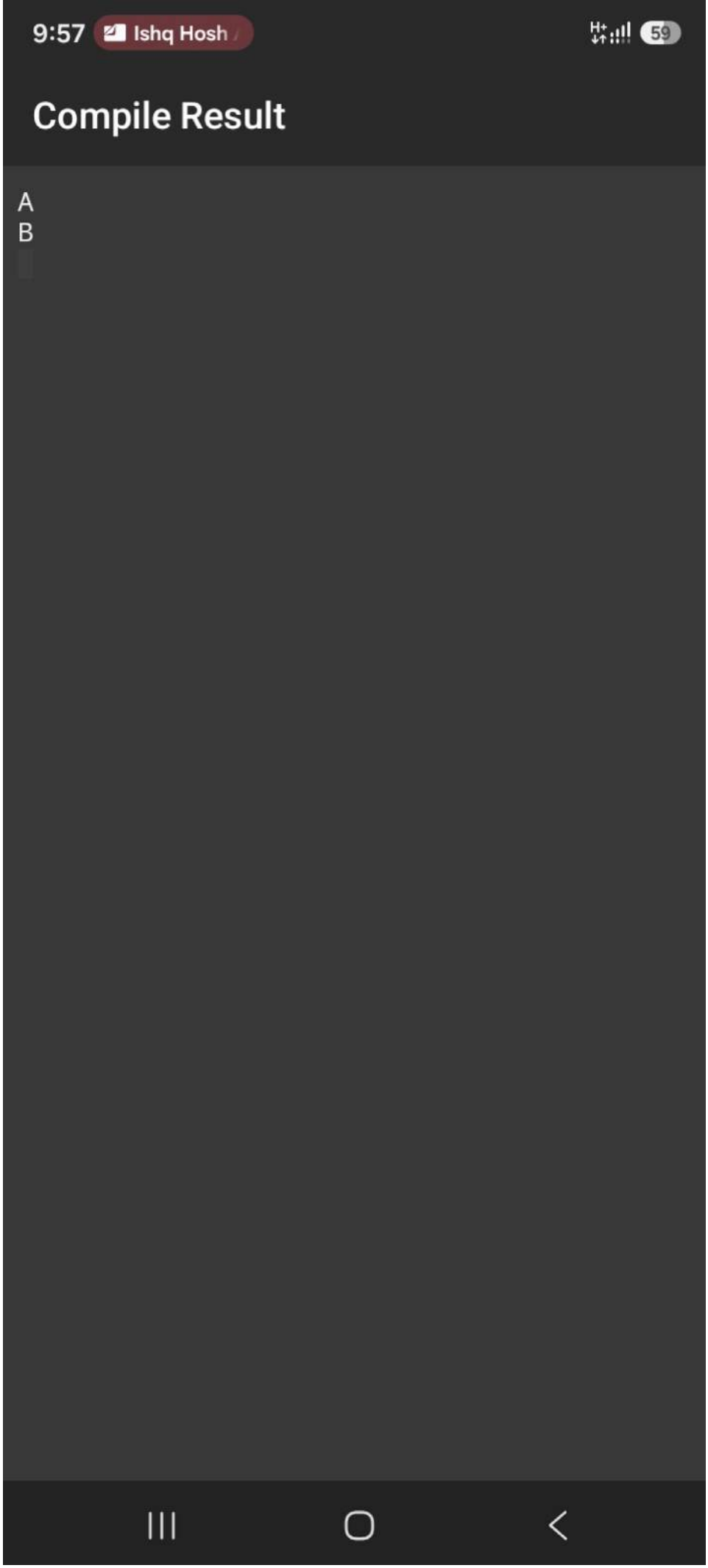
next


9:56
Ishq Hosh
60
Coding C++
Auto saved at 21:56:15
RUN
MENU
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6 char letter;
7
8 letter = 'A';
9 cout << letter << '\n';
10
11 letter = 'B';
12 cout << letter << '\n';
13
14 cin >> letter;
15
16 return 0;
17 }
Tab
{}
“”
;
=
\
&
,

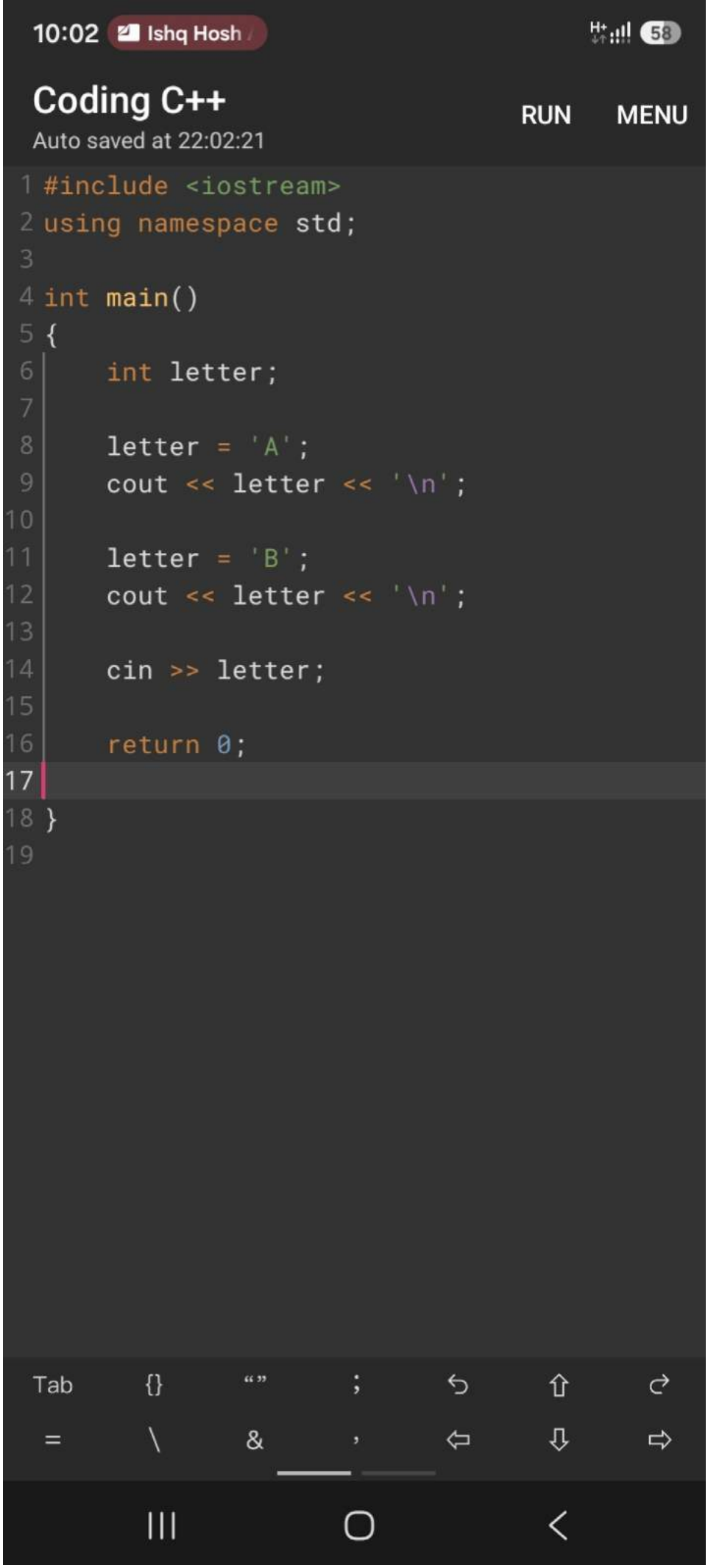
CamScanner



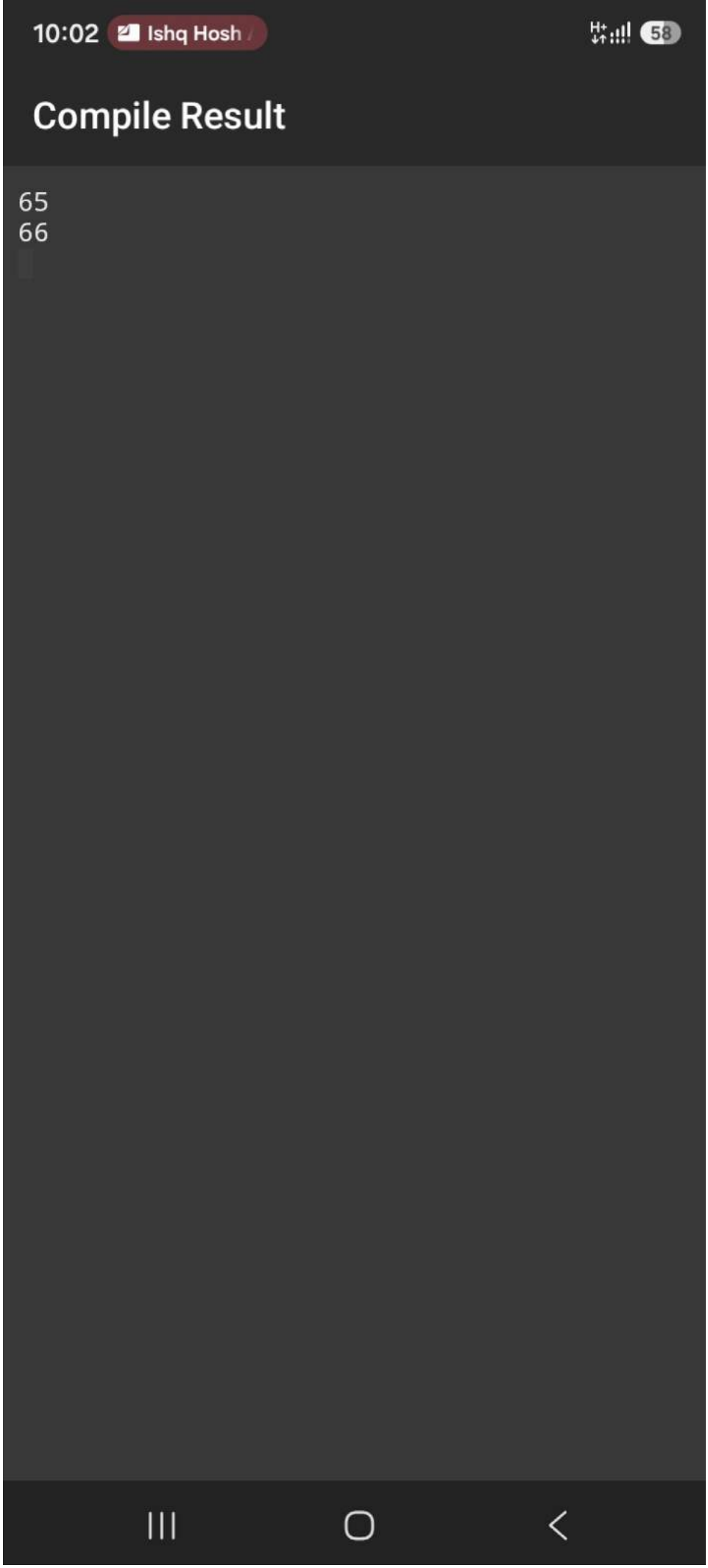
CamScanner



CamScanner


10:02
Ishq Hosh
58
Coding C++
Auto saved at 22:02:21
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
Tab
{}
“”
;
=
\
&
,

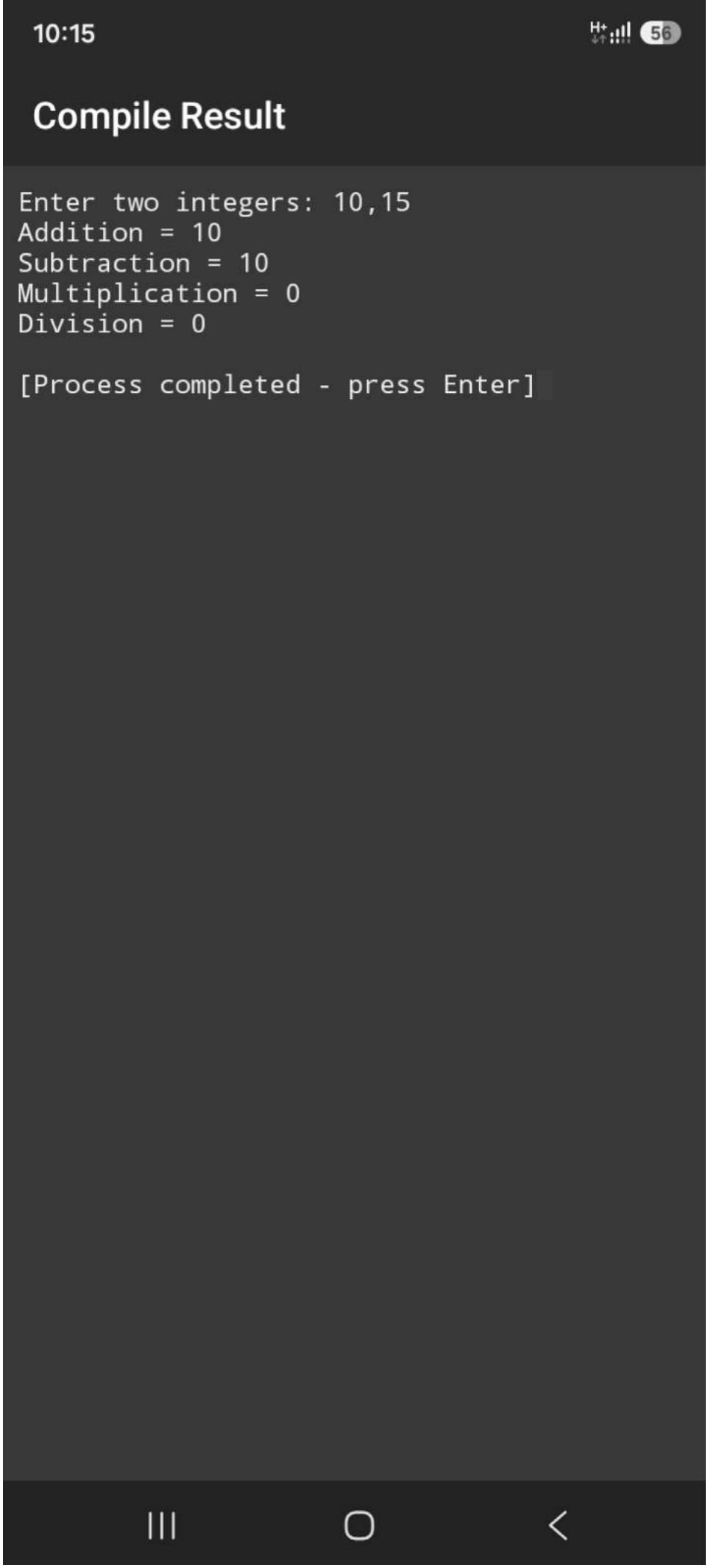
CamScanner



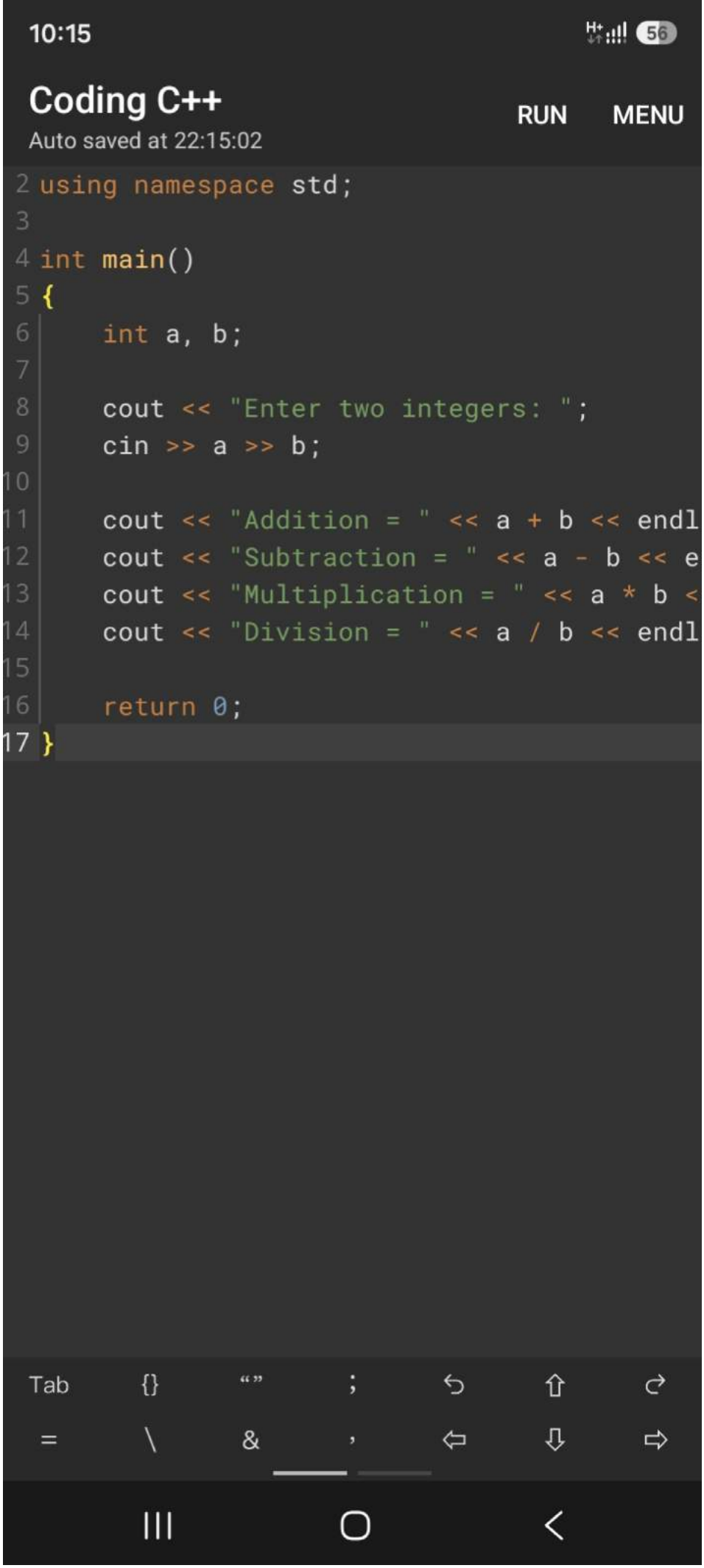
CamScanner


10:11
Ishq Hosh
56
Coding C++
Auto saved at 22:09:40
RUN
MENU
4 int main()
5 {
6 int a, b;
7
8 cout << "Enter two integers: ";
9 cin >> a >> b;
10
11 cout << "Addition = " << a + b << endl
12 cout << "Subtraction = " << a - b << en
13 cout << "Multiplication = " << a * b <<
14 cout << "Division = " << a / b << endl
15
16 return 0;
17
18 }
19
Tab
{}
“”
;
=
\
&
,

CamScanner


10:15
56
Compile Result
Enter two integers: 10,15
Addition = 10
Subtraction = 10
Multiplication = 0
Division = 0
[Process completed - press Enter]

CamScanner


10:15
Coding C++
Auto saved at 22:15:02
RUN
MENU
using namespace std;
int main()
{
int a, b;
cout << "Enter two integers: ";
cin >> a >> b;
cout << "Addition = " << a + b << endl
cout << "Subtraction = " << a - b << e
cout << "Multiplication = " << a * b <
cout << "Division = " << a / b << endl
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

Coding C++

Auto saved at 15:56:03

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;

    cout << "Enter first integer: ";
    cin >> num1;

    cout << "Enter second integer: ";
    cin >> num2;

    cout << "Addition = " << num1 + num2 << enc
    cout << "Subtraction = " << num1 - num2 <<
    cout << "Multiplication = " << num1 * num2
    cout << "Division = " << num1 / num2 << enc

    return 0;
}
```

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Coding C++

Auto saved at 16:09:00

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

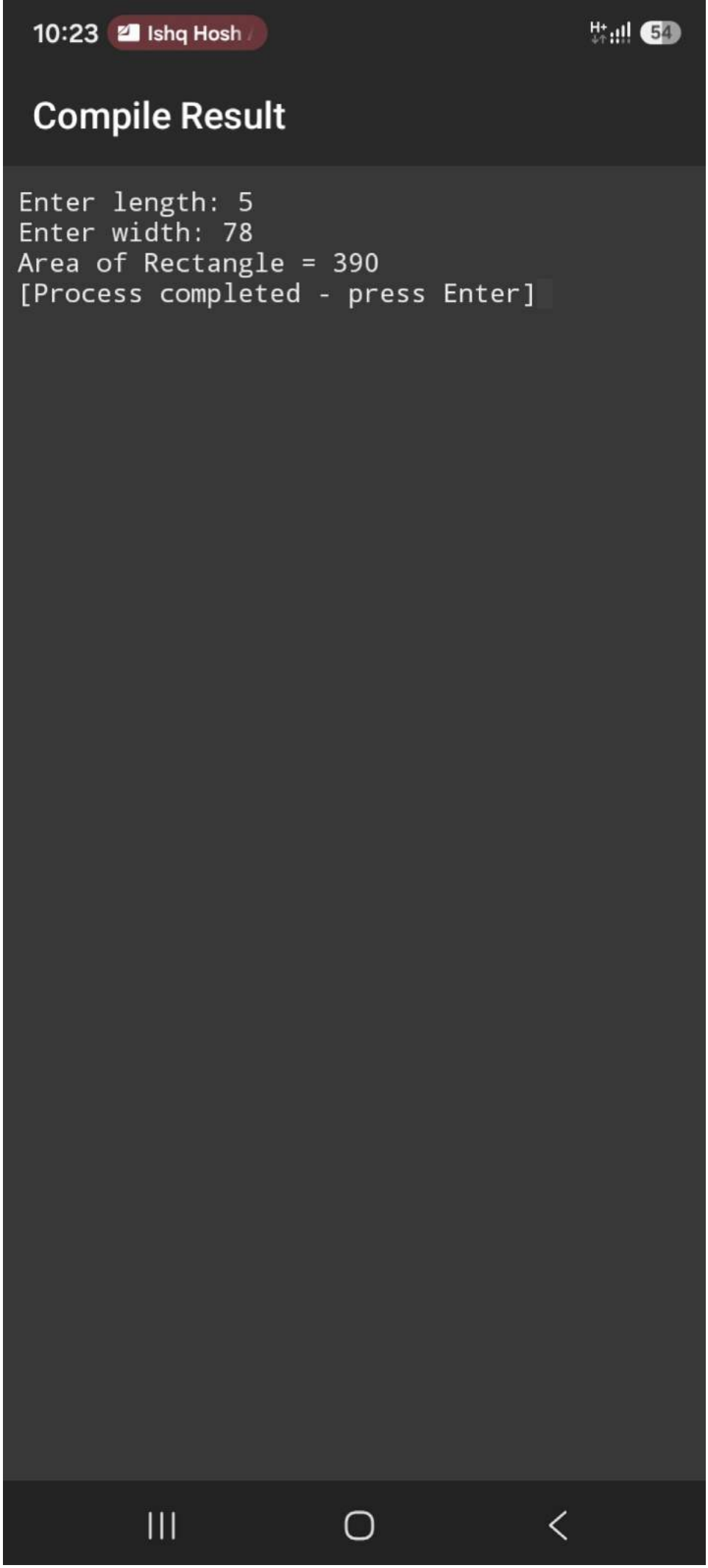
    cout << "Please enter your 3 numbers:";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val

    cout << val3 << "\n" << val2 << "\n" << val

    return 0;
}
```

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10:23
Ishq Hosh
54
Compile Result
Enter length: 5
Enter width: 78
Area of Rectangle = 390
[Process completed - press Enter]

CamScanner

10:23 Ishq Hosh 54

Coding C++

RUN MENU

Auto saved at 22:22:42

```
3
4 int main() {
5     float length, width, area;
6
7     cout << "Enter length: ";
8     cin >> length;
9
10    cout << "Enter width: ";
11    cin >> width;
12
13    area = length * width;
14
15    cout << "Area of Rectangle = " << area
16
17    return 0;
18 }
```

CamScanner

10:24 Ishq Hosh 54

Compile Result

```
Enter length: 5
Enter width: 10
Area of Rectangle = 50
[Process completed - press Enter]
```

CamScanner

10:24 Ishq Hosh 54

Coding C++

Auto saved at 22:24:06

RUN MENU

```
3
4 int main() {
5     float length, width, area;
6
7     cout << "Enter length: ";
8     cin >> length;
9
10    cout << "Enter width: ";
11    cin >> width;
12
13    area = length * width;
14
15    cout << "Area of Rectangle = " << area
16
17    return 0;
18 }
```

CamScanner

10:25 Ishq Hosh

Compile Result

```
Enter base: 5
Enter height: 10
Area of Triangle = 25
[Process completed - press Enter]
```

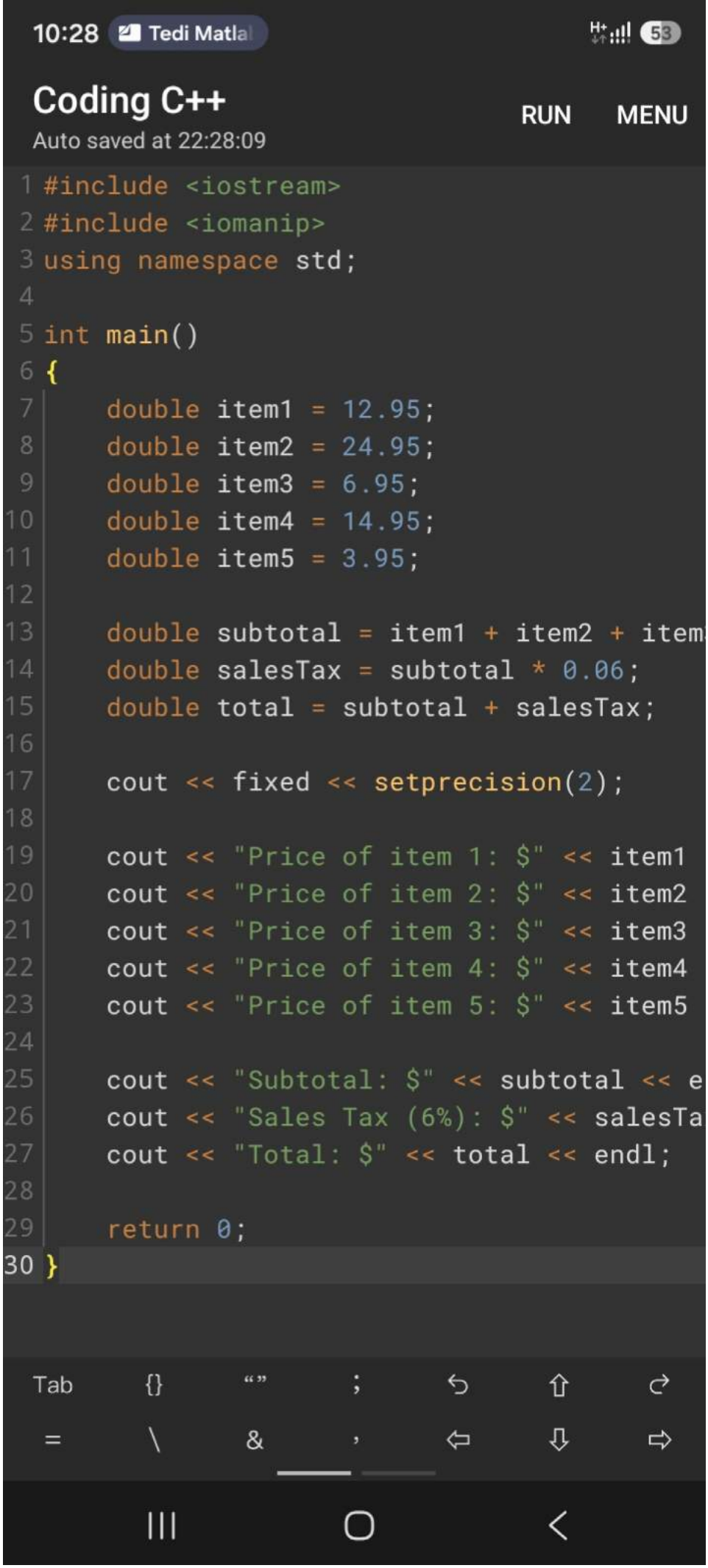
CamScanner


10:25
Ishq Hosh
54
Coding C++
Auto saved at 22:25:11
RUN
MENU
3
4 int main() {
5 float base, height, area;
6
7 cout << "Enter base: ";
8 cin >> base;
9
10 cout << "Enter height: ";
11 cin >> height;
12
13 area = base * height * 0.5;
14
15 cout << "Area of Triangle = " << area;
16
17 return 0;
18 }
Tab
{}
“”
;
=
\
&
,

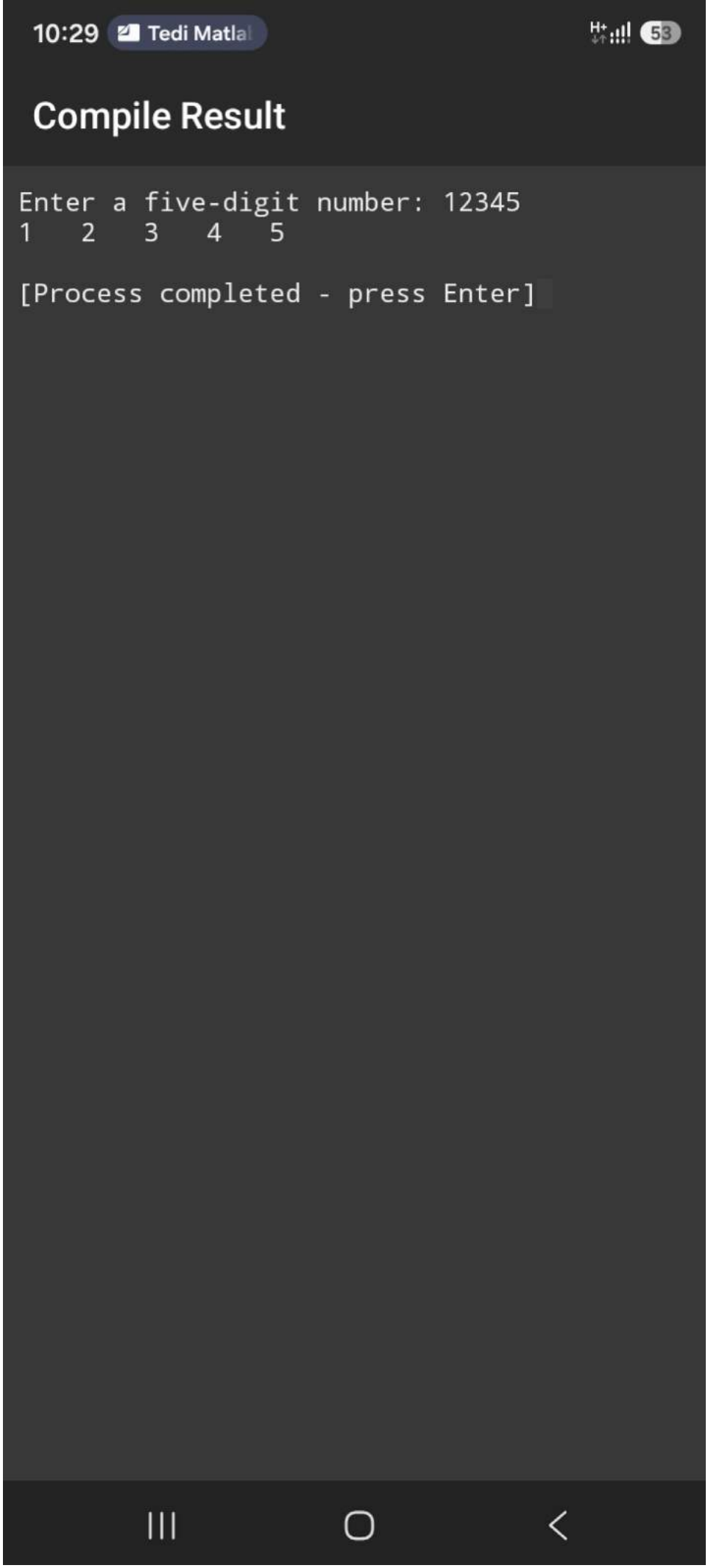
CamScanner


10:28
Tedi Matla
53
Compile Result
Price of item 1: $12.95
Price of item 2: $24.95
Price of item 3: $6.95
Price of item 4: $14.95
Price of item 5: $3.95
Subtotal: $63.75
Sales Tax (6%): $3.82
Total: $67.58
[Process completed - press Enter]

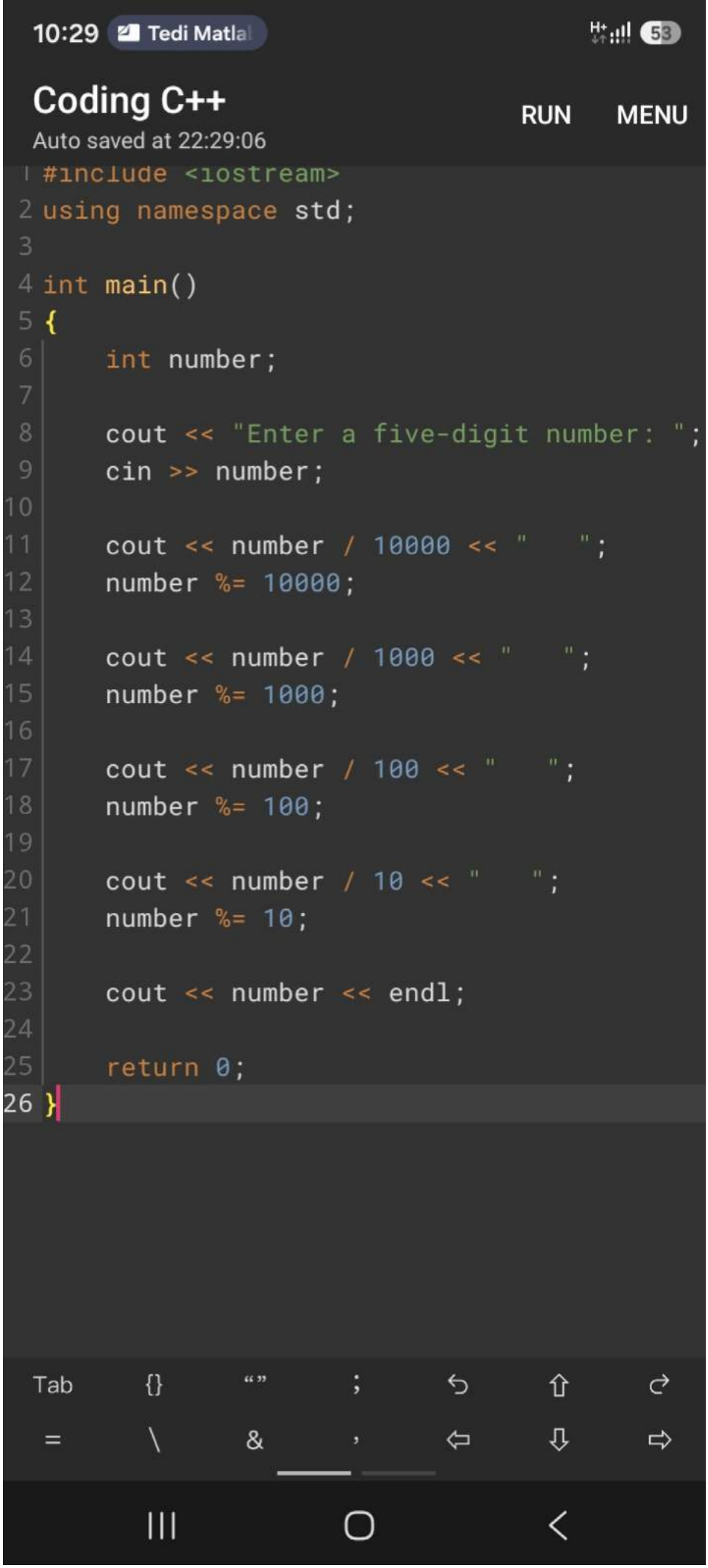
CamScanner


10:28
Tedi Matla
53
Coding C++
Auto saved at 22:28:09
RUN
MENU
1 #include <iostream>
2 #include <iomanip>
3 using namespace std;
4
5 int main()
6 {
7 double item1 = 12.95;
8 double item2 = 24.95;
9 double item3 = 6.95;
10 double item4 = 14.95;
11 double item5 = 3.95;
12
13 double subtotal = item1 + item2 + item
14 double salesTax = subtotal * 0.06;
15 double total = subtotal + salesTax;
16
17 cout << fixed << setprecision(2);
18
19 cout << "Price of item 1: $" << item1
20 cout << "Price of item 2: $" << item2
21 cout << "Price of item 3: $" << item3
22 cout << "Price of item 4: $" << item4
23 cout << "Price of item 5: $" << item5
24
25 cout << "Subtotal: $" << subtotal << e
26 cout << "Sales Tax (6%): $" << salesTa
27 cout << "Total: $" << total << endl;
28
29 return 0;
30 }
Tab
{}
“”
;
=
\
&
,

CamScanner


10:29
Tedi Matla
53
Compile Result
Enter a five-digit number: 12345
1 2 3 4 5
[Process completed - press Enter]

CamScanner


10:29
Tedi Matla
53
Coding C++
Auto saved at 22:29:06
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int number;
cout << "Enter a five-digit number: ";
cin >> number;
cout << number / 10000 << " ";
number %= 10000;
cout << number / 1000 << " ";
number %= 1000;
cout << number / 100 << " ";
number %= 100;
cout << number / 10 << " ";
number %= 10;
cout << number << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

10:32 Tedi Matlal 52

Coding C++

Nothing changed

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int var1, var2, temp;

    cout << "Enter value of var1: ";
    cin >> var1;

    cout << "Enter value of var2: ";
    cin >> var2;

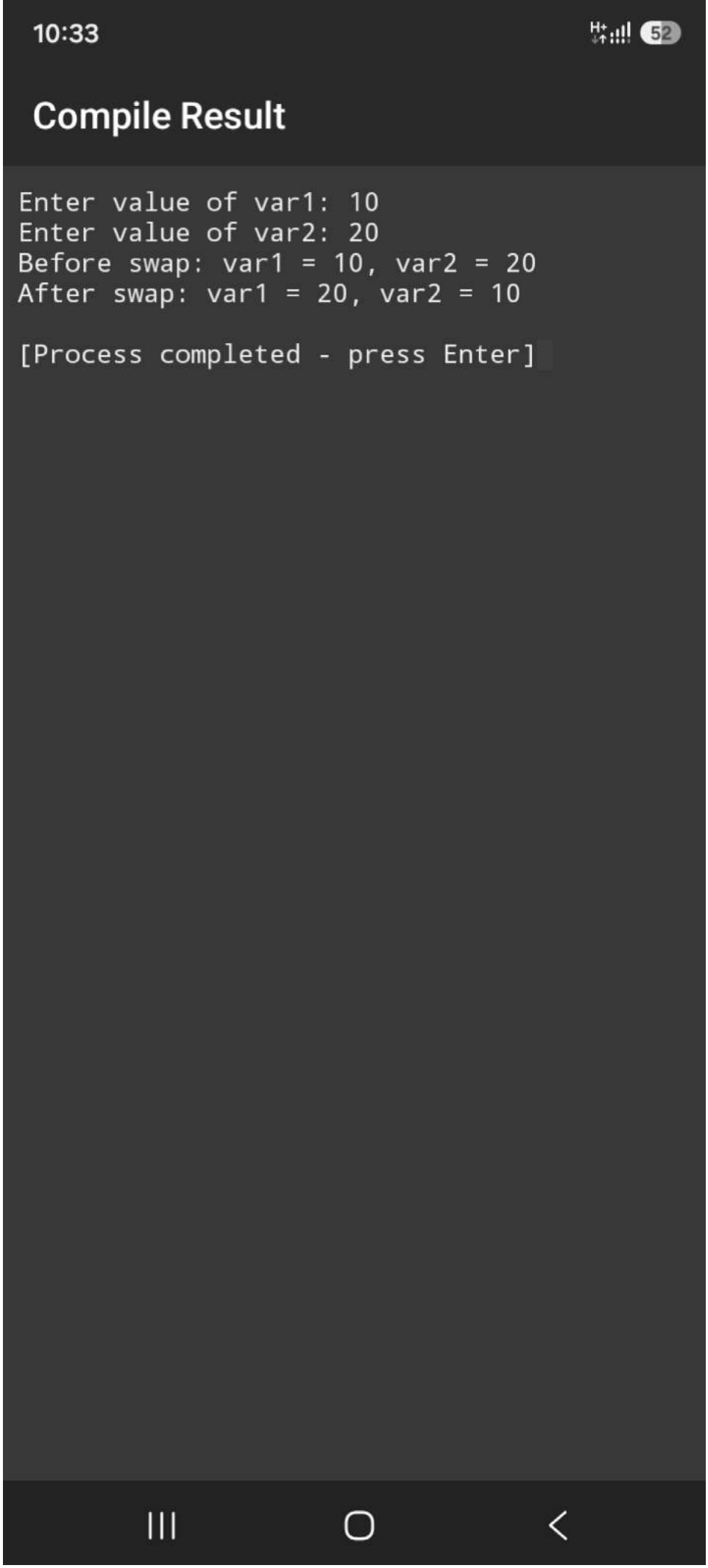
    cout << "Before swap: var1 = " << var1

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After swap: var1 = " << var1

    return 0;
}
```

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10:33
52
Compile Result
Enter value of var1: 10
Enter value of var2: 20
Before swap: var1 = 10, var2 = 20
After swap: var1 = 20, var2 = 10
[Process completed - press Enter]

CamScanner